DAY 23 | PRE-READ

Where GenAI Works, and How to Keep It Safe

A light read before Day 23. About 15 minutes, no code required.

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Why read this?

So far you have built GenAI features. Today you look at where they are used across industries, how organisations grow from experiments to real systems, and how to keep AI agents safe. In the afternoon a security specialist (SME) leads the session and sets an assignment.

1. Same tools, different industries NEW (v4)

A telecom operator, a bank, a shop, a hospital and a factory can all use the same model. What changes is the data, the rules and what happens if the AI is wrong. A wrong product description is embarrassing. A wrong medical note can hurt someone.

Industry	A typical first use case
Telecom	Answering plan and billing questions from approved FAQs
Banking and insurance	Pulling fields out of forms and claims
Retail	Writing product descriptions
Healthcare	Drafting notes for a clinician to check
Manufacturing	Searching maintenance manuals

2. From experiment to everyday tool NEW (v4)

Most AI projects start as an **experiment**, become a **pilot** with a few real users, and only some reach **production**. Later they may be **scaled** across teams, and some become **agentic**, where AI takes actions with tools. Moving up a level needs an owner, proof of value, tests, security checks and a budget.

3. A new kind of colleague needs a badge NEW (v4)

An AI agent that can read tickets and issue refunds is like a new employee. You would give a new employee a badge that opens only the doors they need, for a limited time. Agents need the same: their own **identity**, **scoped permissions** and **short-lived** access.

4. Patching and modernising NEW (v4)

Old software keeps working until a weakness is found in it or in something it depends on. **Patching** fixes known weaknesses quickly and safely. **Modernisation** moves old code to a modern design. Coding agents like OpenAI Codex make both faster, but a person still checks every change and tests

prove it works.

> **Key idea:** the model proposes, the engineer decides. No fix is done until a test proves it.

5. Think about these before class

- Which industry from the table would worry you most if the AI were wrong? Why?
- If an AI agent could act in your company's systems, which three actions should it never take alone?
- Have you ever worked on old code nobody wanted to touch? What made it scary?

Anshin (安心): peace of mind, feeling safe